

PIERO GASPAROTTO

Quantum Software Engineer @ Microsoft Quantum

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📍 Seattle, WA

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EXPERIENCE

Senior Quantum Application Engineer

Quantum Applications, Microsoft Quantum

📅 Oct 2023 – Present

📍 Redmond, USA

I lead the development of end-to-end modelling solutions at the interface of machine-learning force fields, materials science, computational chemistry, quantum computing, and cloud-native agentic workflows, working closely with research teams, strategic partners, and customers to translate scientific requirements into production-ready solutions.

Principal ML Scientist

AI4Chemistry, Ro5

📅 Mar 2023 – Oct 2023

📍 New York, USA

Drove development of Ro5's 3D structure-based drug design platform and co-authored a peer-reviewed paper introducing Hydrascreen, a novel ML model for predicting protein-ligand binding affinities

Software Scientist

Scientific IT, Paul Scherrer Institute

📅 May 2021 – Oct 2023

📍 Villigen, CH

Developed and deployed machine-learning methods for real-time feedback during kHz serial X-ray protein crystallography experiments, collaborating closely with beamline scientists and users.

Postdoc

Empa – nanotech@surfaces Group

📅 Apr 2019 – Mar 2021

📍 Dübendorf, CH

Focused on ML Force Fields, molecular dynamics and enhanced-sampling workflows, super-hard alloys and modeling of surface-catalysis reactions.

Postdoc

University College London – ICE Group

📅 Apr 2018 – Mar 2019

📍 London, UK

Developed a novel machine-learning force field for carbon-based materials and studied glassy interfacial water properties by combining enhanced-sampling with molecular dynamics.

Ph.D Researcher

EPFL – COSMO Group

📅 Oct 2013 – Mar 2018

📍 Lausanne, CH

My research focused on unsupervised learning and pattern recognition methods combined with molecular dynamics and enhanced sampling techniques, with application to liquids, proteins and molecular crystals.

SELECTED PUBLICATIONS

- **QDK/Chemistry**: A Modular Toolkit for Quantum Chemistry Applications *arXiv preprint arXiv:2601.15253* (2026)
- **SimPoly**: Simulation of Polymers with Machine Learning Force Fields Derived from First Principles *arXiv preprint arXiv:2510.13696* (2025)

Full list: Google Scholar

EDUCATION

Ph.D Computational Modelling

EPFL

📅 2013–2018

📍 Lausanne, CH

MS Materials Science

University of Padua

📅 2009–2012

📍 Padova, IT

BS Materials Science

University of Padua

📅 2006–2009

📍 Padova, IT

FUNDING

🏆 Torres-Quevedo Technological Transfer Grant (2019)

With Acellera Labs, for hybrid Quantum/ML approaches for modeling metallo-proteins (€98k)

🖥️ HPC Compute Grants

Awarded 5 compute grants at CSCS and UKCP/MCC (\$350k)

TECH SKILLS

- PyTorch
- Python
- Scripting
- Quantum Chemistry
- Molecular Dynamics
- MLFFs
- Agentic AI
- Deep Learning
- Cloud Computing
- Distributed Systems
- Scientific Software
- Workflow Orchestration
- Customer-facing Applications

MANAGEMENT

- Project Lead
- Agile
- Mentorship
- Cross-functional Collaboration

OPEN SOURCE

QDK/Chemistry — Microsoft Quantum Applications Toolkit

TORO-Indexer — A PyTorch indexer for SX crystallography

Hydrascreen — Protein-ligand binding affinities

PLUMED — Enhanced sampling for molecular dynamics

PAMM — Probabilistic analysis of molecular motifs

LANGUAGES

Italian
English
French

